Causal Discovery

STSCI/INFO/ILRST 3900: Causal Inference

Oct 6 2026

Logistics

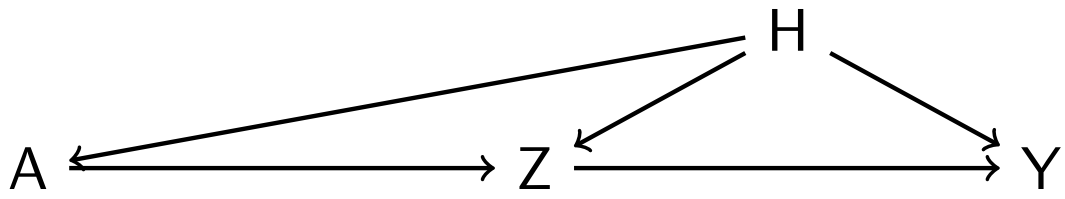
- Quiz 2 on Thursday in class (accommodations through ATP)
- Bring pencil/pen
- Review practice questions in discussion section Wednesday

FAQs

Causal paths are paths that follow the direction of the arrows

- Does not depend on conditioning set
- Different characteristic than open/closed (i.e., can be open causal, open non-causal, blocked causal, blocked non-causal)

Practice



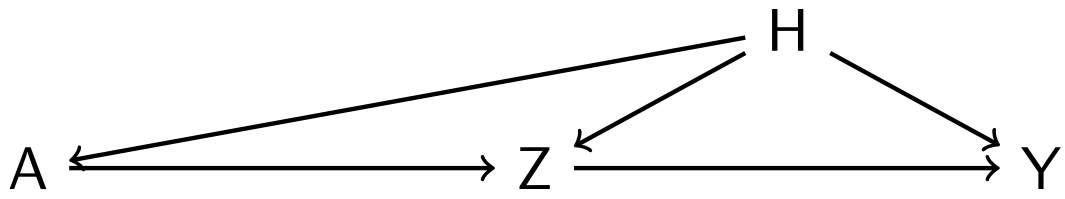
- $A \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$ causal path
- $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$ non-causal
- $A \rightarrow \underbrace{Z}_{\text{Col}} \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$ non-causal
- $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$ non-causal

FAQs

Conditional Exchangeability depends on checking the entire graph, not just individual paths

- Conditional exchangeability holds if: all open paths are causal paths
- Conditional exchangeability does not hold if: any open paths are non-causal

Practice



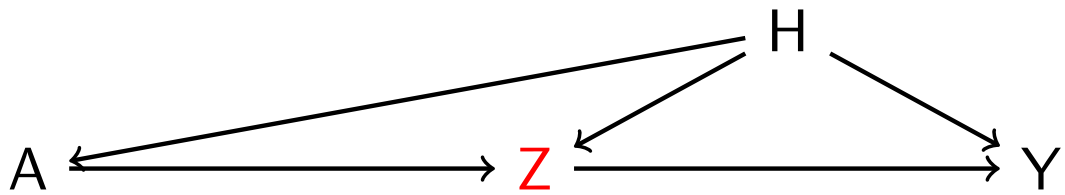
When conditioning on H:

- $A \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$ causal and open
- $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$ non-causal and blocked
- $A \rightarrow \underbrace{Z}_{\text{CoI}} \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$ non-causal and blocked
- $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$ non-causal and blocked

Conditional exchangeability holds when conditioning on H

FAQs

A specific node can be a collider on one path and a non-collider on another



- $A \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$
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Causal Discovery

- So far, we have assumed the DAG is known from expert knowledge
- DAG tells us about conditional independence we would observe in data

$$\text{DAG} \Rightarrow \text{Conditional independence in data}$$

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- DAG tells us about conditional independence we would observe in data

$$\text{DAG} \Rightarrow \text{Conditional independence in data}$$

- Conditional independence is a observational quantity (i.e., not causal)
- Can be tested in observed data
- Can we go in the opposite direction?

$$\text{Conditional independence in data} \overset{?}{\Rightarrow} \text{DAG}$$

Causal Discovery: biological systems

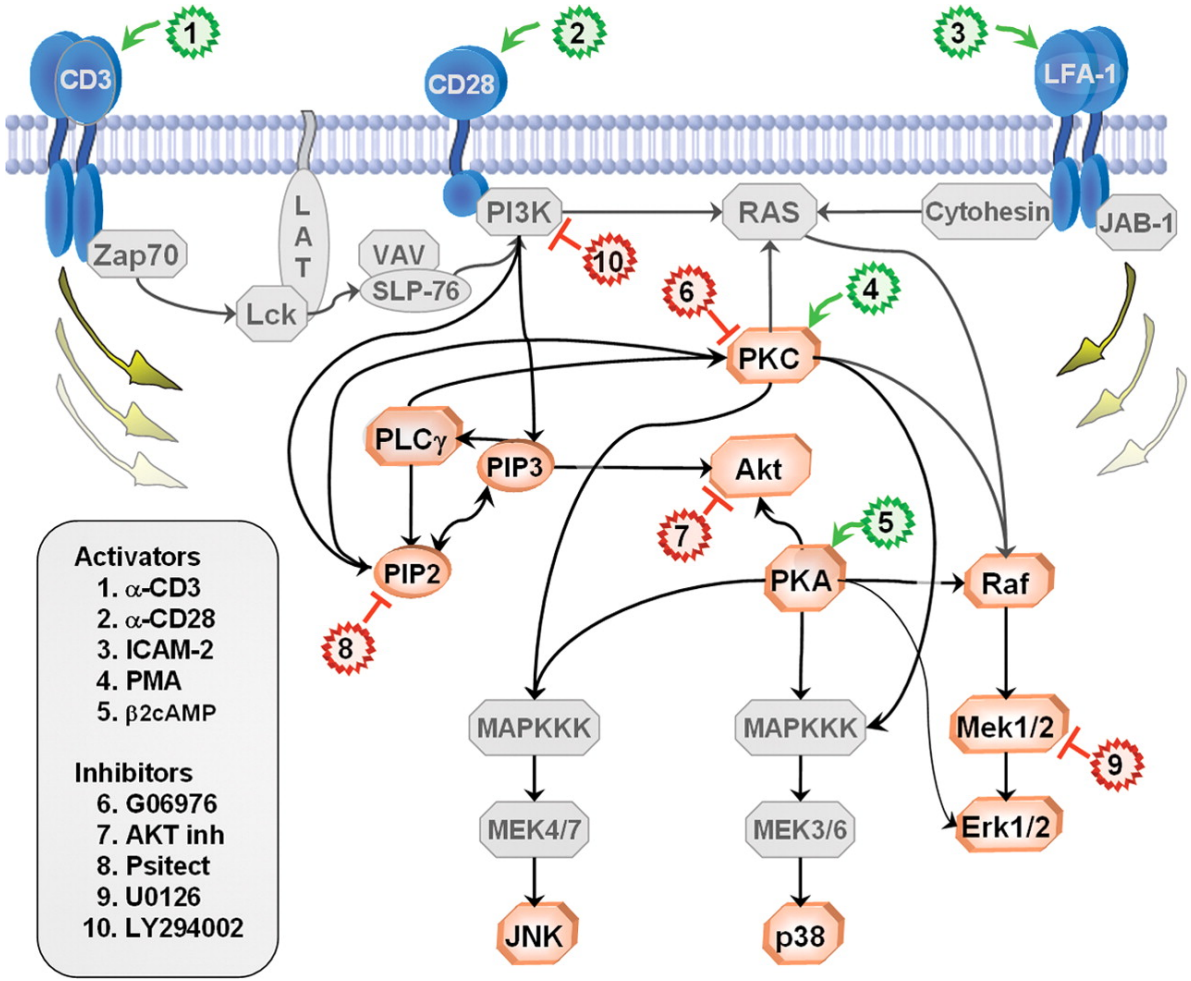


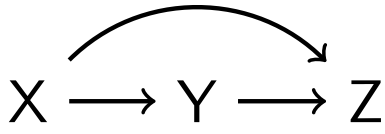
Figure: Regulatory network from Sachs et al (2005, Science)

Causal Discovery

Can we tell which nodes are/aren't connected by an edge?

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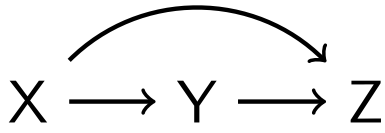
- $X \perp\!\!\!\perp Y$? No
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If there is an edge between two nodes, they cannot be made conditionally independent!

Rule 1

- Start with (undirected) edges between every pair of nodes
- If you can find a set L such that $X \perp\!\!\!\perp Y \mid L$, take away the edge between X and Y

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Allows us to find where the edges are, but not necessarily direction

A **skeleton** is the DAG where we have made all edges undirected

$$\text{DAG} : X \to Y \to Z \qquad \text{Skeleton} : X - Y - Z$$

Causal Discovery

Can we also tell which direction an edge points?

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$X \longrightarrow Y \longrightarrow Z$

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Colliders can sometimes tell us the direction of an edge

Rule 2

- Suppose we have $X - Y - Z$ and no edge between X and Z
- Suppose $X \not\perp\!\!\!\perp Y \mid L$ for some set L that does not contain Y

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- Then, $X \rightarrow Y \leftarrow Z$

Rule 2

- Suppose we have $X - Y - Z$ and no edge between X and Z
- Suppose $X \not\perp\!\!\!\perp Y \mid L$ for some set L that does not contain Y
- Then, $X \to Y \leftarrow Z$
- **Unshielded collider**: $X \to Y \leftarrow Z$ and X and Z do not have an edge

Causal Discovery

Can we fully determine the graph from data?

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Some graphs have the exact same set of conditional independence statements and cannot be distinguished from data alone!

Rule 3

Graphs imply the same conditional independence statements if

- Same skeleton: edges in the same location, but possibly different direction (from Rule 1)
- Same unshielded colliders: $X \rightarrow Y \leftarrow Z$ and X and Z do not share an edge (from Rule 2)